NOTES 06: TWO QUANTITATIVE VARIABLES

Stat 120 | Spring 2026
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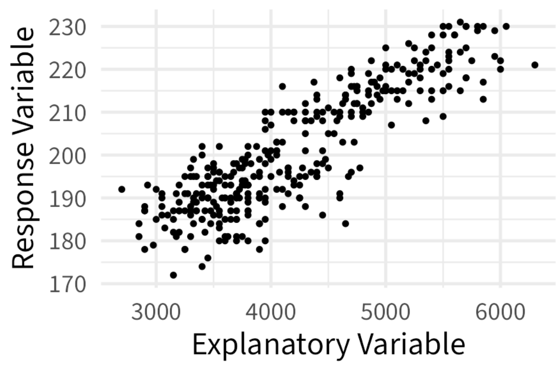
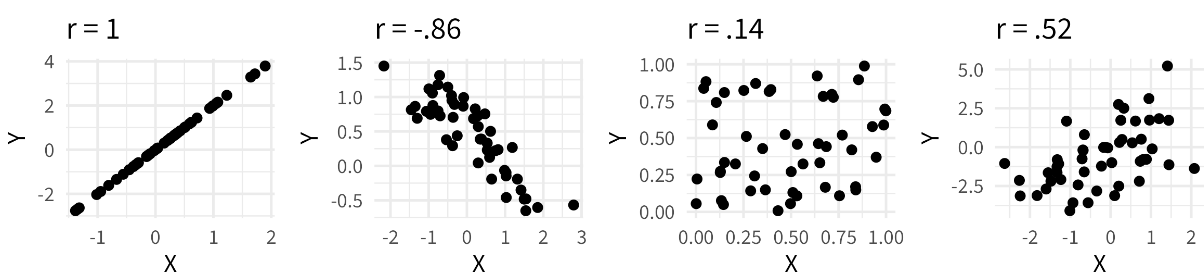
When we're interested in the relationship between two quantitative variables, the best visualization is a **scatterplot**.



If we want to summarize the relationship in a single number, we'll often choose the **correlation**.

Correlation: measure of the strength and direction of a linear relationship between two quantitative variables

- sample: population:
- between -1 and +1
- no units, doesn't matter what x and y are $r = \frac{1}{n-1} \sum (\frac{(x_i - \bar{x})}{s_x})(\frac{(y_i - \bar{y})}{s_y})$

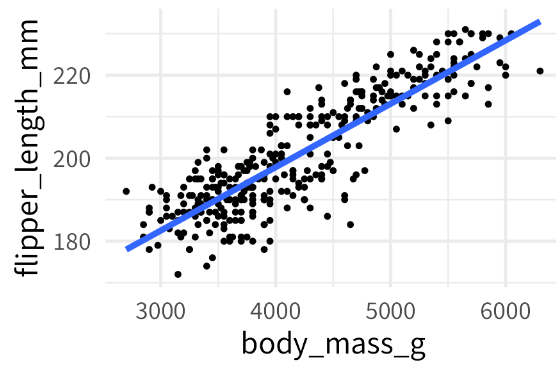



```
cor(penguins$body_mass_g, penguins$flipper_length_mm, use = "complete.obs")
```

```
[1] 0.8712
```

If the relationship is **linear**, we can also summarize the relationship with “the line of best fit” or “least squares” line.

```r
ggplot(penguins, aes(x = body_mass_g, y = flipper_length_mm)) +
  geom_point(size = .5) +
  geom_smooth(method = "lm", se = FALSE)
```




```r
lm(flipper_length_mm ~ body_mass_g, data = penguins)
```

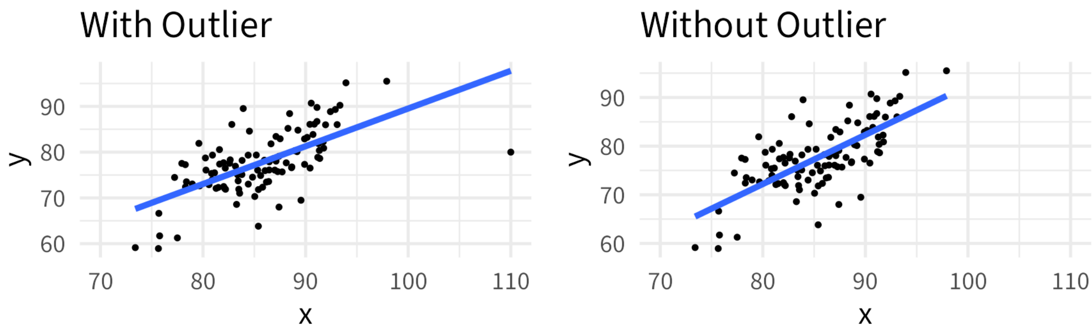
```
(Intercept) body_mass_g
  136.72956     0.01528
```

$$\widehat{\text{flipper\_length\_mm}} = 136.73 + 0.02(\text{body\_mass\_g}) \tag{1}$$

Interpretations:

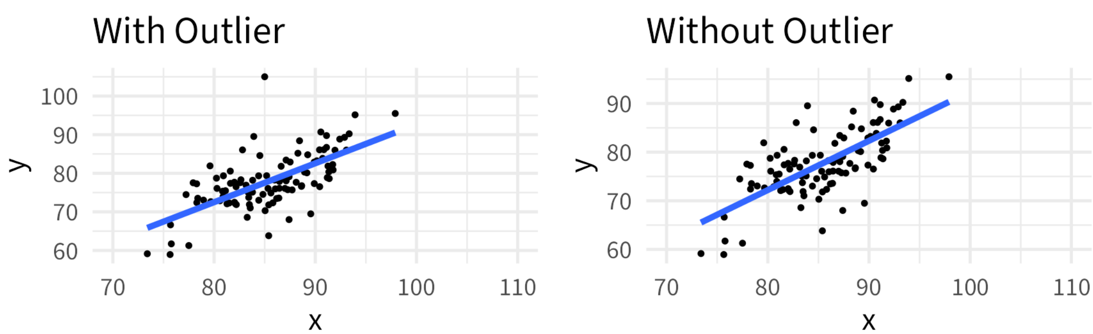
- **Slope:** As [x variable] increases by 1 [unit], the average [y variable] also increases by [b1]
- **Intercept:** For [observation with x = 0], the model predicts the average [y variable] to be [b0]

Two types of outliers:



Best fit line with outlier: $\hat{y} = 7.1 + 0.82$ x

Best fit line without outlier: $\hat{y} = -8.79 + 1.01$ x



Best fit line with outlier: $\hat{y} = -8.1 + 1.01$ x

Best fit line without outlier: $\hat{y} = -8.79 + 1.01$ x